

Mathletes Formulas

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Table of Contents

1. Polynomials	2
1.1. Linear Equations	2
1.2. Quadratics	2
1.3. General Polynomials	2
1.4. Rational Functions	3
2. Trigonometry	3
2.1. Functions	3
2.2. Identities	5
2.3. Laws	6
3. Graphs and Geometry	6
3.1. Lines and Points	6
3.2. Shapes	6
3.3. Graph Manipulation	7
4. Miscellaneous	7
4.1. Inverse Functions	7
4.2. Complex Numbers	7
4.3. Exponential and Logarithmic Properties	8

1. Polynomials

1.1. Linear Equations

Slope-Intercept	$y = mx + b$	Slope	$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$
Point-Slope	$y - y_1 = m(x - x_1)$	Point-Point	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Parallel lines have equivalent slopes, while perpendicular lines have slopes that are negative reciprocals of one another ($m \perp -\frac{1}{m}$)

1.2. Quadratics

Quadratic Equations

Standard Form $ax^2 + bx + c$

Let $D = b^2 - 4ac$,

Vertex Form $a(x - h)^2 + k$

1. $D > 0 \implies 2$ real solutions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. $D = 0 \implies 1$ real solution

3. $D < 0 \implies$ no real solutions

Minimum and Maximum Values

Consider the function $f(x) = ax^2 + bx + c$ with vertex $(-\frac{b}{2a}, f(-\frac{b}{2a}))$

1. $a > 0 \implies f$ has a *minimum* at $x = -\frac{b}{2a}$. The minimum value is $f(-\frac{b}{2a})$.
2. $a < 0 \implies f$ has a *maximum* at $x = -\frac{b}{2a}$. The maximum value is $f(-\frac{b}{2a})$.

1.3. General Polynomials

Let n be a non-negative integer and let $a_n, a_{n-1}, \dots, a_2, a_1, a_0$ be real numbers such that $a_n \neq 0$. The function:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

is a polynomial function of x with degree n .

Linear Factorization Theorem

If $f(x)$ is a polynomial of degree n , where $n > 0$, then $f(x)$ has precisely n linear factors

$$f(x) = a_n(x - c_1)(x - c_2)\dots(x - c_n)$$

where $c_1, c_2, \dots, c_n$ are complex numbers.

Rational Zero Test

If the polynomial has *integer* coefficients, then every rational zero of f has the form $\frac{p}{q}$, where p and q have no common factors other than 1, and

p = a factor of the constant term a_0

q = a factor of the leading coefficient a_n

Conjugate Pairs

Let f be a polynomial function with *real coefficients*. If $a + bi$, where $b \neq 0$, is zero of f , the complex conjugate $a - bi$ is also a zero of f .

For polynomials with *integer coefficients*, $a + \sqrt{b}$ being a zero implies $a - \sqrt{b}$ is also a zero.

Roots

Let f be the polynomial

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

$$\text{The sum of all roots} = -\frac{a_n}{a_{n-1}}$$

$$\text{The product of all roots} = (-1)^n \frac{a_0}{a_n}$$

1.4. Rational Functions

Let f be the rational function

$$f(x) = \frac{N(x)}{D(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + \dots + b_1 x + b_0}$$

where $N(x)$ and $D(x)$ have no common factors.

1. The graph of f has *vertical asymptotes* at the zeros of $D(x)$.
2. The graph of f has at most one *horizontal asymptote* determined by comparing the degrees of $N(x)$ and $D(x)$.
 - a. $n < 0 \implies$ the line $y = 0$ is the horizontal asymptote.
 - b. $n = 0 \implies$ the line $y = \frac{a_n}{b_m}$ is the horizontal asymptote.
 - c. $n > 0 \implies$ there is no horizontal asymptote.

Suppose $N(x)$ and $D(x)$ had some common factor $(x - c)$. The graph of f would then have a hole at $(c, f(c))$

2. Trigonometry

2.1. Functions

$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$	$\csc \theta = \frac{\text{hypotenuse}}{\text{opposite}}$	$\sin \theta = \frac{y}{1} = y$	$\csc \theta = \frac{1}{y}$
$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$	$\sec \theta = \frac{\text{hypotenuse}}{\text{adjacent}}$	$\cos \theta = \frac{x}{1} = x$	$\sec \theta = \frac{1}{x}$
$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$	$\cot \theta = \frac{\text{adjacent}}{\text{opposite}}$	$\tan \theta = \frac{y}{x}$	$\cot \theta = \frac{x}{y}$

Function	Domain	Range	Period
sin	$\mathbb{R}$	$[-1, 1]$	2π
cos	$\mathbb{R}$	$[-1, 1]$	2π
tan	$\{x \mid x \in \mathbb{R}, (x + \frac{\pi}{2}) \bmod \pi \neq 0\}$	$\mathbb{R}$	π
csc	$\{x \mid x \in \mathbb{R}, x \bmod \pi \neq 0\}$	$(-\infty, -1] \cup [1, \infty)$	2π
sec	$\{x \mid x \in \mathbb{R}, (x + \frac{\pi}{2}) \bmod \pi \neq 0\}$	$(-\infty, -1] \cup [1, \infty)$	2π
cot	$\{x \mid x \in \mathbb{R}, x \bmod \pi \neq 0\}$	$\mathbb{R}$	π

Table 1: Functions

Function	Domain	Range
$\sin^{-1} / \arcsin$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} / \arccos$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} / \arctan$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\csc^{-1} / \operatorname{arccsc}$	$(-\infty, -1] \cup [1, \infty)$	$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$
$\sec^{-1} / \operatorname{arcsec}$	$(-\infty, -1] \cup [1, \infty)$	$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$
$\cot^{-1} / \operatorname{arccot}$	$\mathbb{R}$	$(0, \pi)$

Table 2: Inverse Functions

	0°	30°	45°	60°	90°	180°	270°	360°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0	1
tan	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undef	0	undef	0

Table 3: Common Values

2.2. Identities

Radians and Degrees

$$n^\circ = \frac{\pi}{180}n \text{ rad}$$

$$m \text{ rad} = \left(\frac{180}{\pi}m\right)^\circ$$

Reciprocals

$$\begin{aligned} \csc \theta &= (\sin \theta)^{-1} & \sin \theta &= (\csc \theta)^{-1} \\ \sec \theta &= (\cos \theta)^{-1} & \cos \theta &= (\sec \theta)^{-1} \\ \cot \theta &= (\tan \theta)^{-1} & \tan \theta &= (\cot \theta)^{-1} \end{aligned}$$

Pythagorean

$$\begin{aligned} \sin^2 \theta + \cos^2 \theta &= 1 \\ \tan^2 \theta + 1 &= \sec^2 \theta \\ 1 + \cot^2 \theta &= \csc^2 \theta \end{aligned}$$

Even/Odd

$$\begin{aligned} \sin(-\theta) &= -\sin \theta & \csc(-\theta) &= -\csc \theta \\ \cos(-\theta) &= \cos \theta & \sec(-\theta) &= \sec \theta \\ \tan(-\theta) &= -\tan \theta & \cot(-\theta) &= -\cot \theta \end{aligned}$$

Sum-to-Difference

$$\begin{aligned} \sin(\alpha \pm \beta) &= \sin \alpha \cos \beta \pm \cos \alpha \sin \beta \\ \cos(\alpha \pm \beta) &= \cos \alpha \cos \beta \mp \sin \alpha \sin \beta \\ \tan(\alpha \pm \beta) &= \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta} \end{aligned}$$

Half-Angle

$$\begin{aligned} \sin\left(\frac{\theta}{2}\right) &= \pm \sqrt{\frac{1 - \cos \theta}{2}} \\ \cos\left(\frac{\theta}{2}\right) &= \pm \sqrt{\frac{1 + \cos \theta}{2}} \end{aligned}$$

$$\tan\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \frac{\sin \theta}{1 + \cos \theta} = \frac{1 - \cos \theta}{\sin \theta}$$

Double-Angle

$$\begin{aligned} \sin 2\theta &= 2 \sin \theta \cos \theta \\ \cos 2\theta &= \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta \\ \tan 2\theta &= \frac{2 \tan \theta}{1 - \tan^2 \theta} \end{aligned}$$

Sum-to-Product

$$\begin{aligned} \sin \alpha + \sin \beta &= 2 \sin\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) \\ \sin \alpha - \sin \beta &= 2 \cos\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right) \\ \cos \alpha + \cos \beta &= 2 \cos\left(\frac{\alpha + \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) \\ \cos \alpha - \cos \beta &= -2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right) \end{aligned}$$

$$\begin{aligned} \tan \alpha \pm \tan \beta &= \frac{\sin(\alpha \pm \beta)}{\cos \alpha \cos \beta}, \\ \cos \alpha \cos \beta &\neq 0 \\ \cot \alpha \pm \cot \beta &= \frac{\sin(\beta \pm \alpha)}{\sin \alpha \sin \beta}, \\ \sin \alpha \sin \beta &\neq 0 \end{aligned}$$

Product-to-Sum

$$\begin{aligned} \sin \alpha \sin \beta &= \frac{\cos(\alpha - \beta) - \cos(\alpha + \beta)}{2} \\ \cos \alpha \cos \beta &= \frac{\cos(\alpha + \beta) + \cos(\alpha - \beta)}{2} \\ \sin \alpha \cos \beta &= \frac{\sin(\alpha + \beta) + \sin(\alpha - \beta)}{2} \end{aligned}$$

Co-Function

$$\begin{aligned} \sin(90^\circ - \theta) &= \cos \theta & \cos(90^\circ - \theta) &= \sin \theta \\ \sec(90^\circ - \theta) &= \csc \theta & \csc(90^\circ - \theta) &= \sec \theta \\ \tan(90^\circ - \theta) &= \cot \theta & \cot(90^\circ - \theta) &= \tan \theta \end{aligned}$$

2.3. Laws

Law of Sines

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

$$\frac{\sin \alpha}{a} = \frac{\sin \beta}{b} = \frac{\sin \gamma}{c}$$

Law of Cosines

$$a^2 = b^2 + c^2 - 2bc \cos \alpha$$

$$\cos \alpha = \frac{b^2 + c^2 - a^2}{2bc}$$

Law of Tangents

$$\frac{a-b}{a+b} = \frac{\tan\left(\frac{1}{2}(\alpha-\beta)\right)}{\tan\left(\frac{1}{2}(\alpha+\beta)\right)}$$

3. Graphs and Geometry

3.1. Lines and Points

Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Pythagorean Theorem

$$a^2 + b^2 = c^2 \implies c = \sqrt{a^2 + b^2}$$

Perpendicular Bisector

$$2(x_2 - x_1)x + 2(y_2 - y_1)y = x_2^2 + y_2^2 - x_1^2 - y_1^2$$

3.2. Shapes

Circles

$$(x - m)^2 + (y - n)^2 = r^2$$

$$\text{Area} = \pi r^2$$

$$\text{Circumference} = 2\pi r$$

Regular Polygons

$$\text{Sum of Interior Angles} = 180(n - 2)^\circ$$

$$\text{Area} = \frac{1}{2} \cdot \text{Apothem} \cdot \text{Perimeter}$$

Volumes

$$V_{\text{Cylinder}} = \pi r^2 h$$

$$V_{\text{Cone}} = \frac{1}{3} \pi r^2 h$$

$$V_{\text{Sphere}} = \frac{4}{3} \pi r^3$$

$$V_{\text{Pyramid}} = \frac{1}{3} B h$$

Triangle Similarity

Two triangles $\triangle ABC$ and $\triangle DEF$ are similar if

SSS

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

SAS

$$\frac{AB}{DE} = \frac{AC}{DF} \text{ and } m\angle A = m\angle D$$

AA

$$m\angle A = m\angle D \\ m\angle B = m\angle E$$

3.3. Graph Manipulation

Transformation	Conditions	Description
$g(x) = f(x \pm c)$	(+) (-)	shift left c units shift right c units
$g(x) = f(x) \pm c$	(+) (-)	shift left c units shift right c units
$g(x) = cf(x)$	$c < 1$ $c > 1$	vertical compress by a factor of c vertical stretch by a factor of c
$g(x) = f(cx)$	$c < 1$ $c > 1$	horizontal stretch by a factor of c horizontal compress by a factor of c
$g(x) = -f(x)$	N/A	reflect about the x-axis
$g(x) = f(-x)$	N/A	reflect about the y-axis

4. Miscellaneous

4.1. Inverse Functions

Let f and g be two function such that

$$f(g(x)) = x \text{ for all } x \text{ in the domain of } g \qquad g(f(x)) = x \text{ for all } x \text{ in the domain of } f$$

Under these conditions, the function g is the *inverse function* of the function f (and vice-versa). The function g can then be denoted as f^{-1} . So,

$$f(f^{-1}(x)) = x \qquad f^{-1}(f(x)) = x$$

The domain of f must be equal to the range of f^{-1} , and the range of f must be equal to the domain of f^{-1} .

4.2. Complex Numbers

Let a and b be real numbers. The number $a + bi$ is a *complex number* written in *standard form*. The number a is the *real part* and the number bi is the *imaginary part* of the complex number.

When $b = 0$, the number $a + bi$ is a real number. When $b \neq 0$, the number $a + bi$ is an *imaginary number*. A number of the form bi , where $b \neq 0$, is a *pure imaginary number*.

$$i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1, i^n = i^{n \bmod 4}$$

4.3. Exponential and Logarithmic Properties

Exponential

$$\begin{array}{ll}
 a^0 = 1 & \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \\
 a^m \cdot a^n = a^{m+n} & (a^m)^n = a^{m \cdot n} \\
 \frac{a^m}{a^n} = a^{m-n} & a^{\frac{1}{n}} = \sqrt[n]{a} \\
 (ab)^n = a^n \cdot b^n & a^m = a^n \implies m = n
 \end{array}$$

Logarithmic

$$\begin{array}{ll}
 \log_n 1 = 0 & \log_n a^b = b \log_n a \\
 \log_n n = 1 & \log_{n^b} a = \frac{1}{b} \log_n a \\
 \log x = \log_{10} x & \log_n ab = \log_n a + \log_n b \\
 \ln x = \log_e x & \log_n \frac{a}{b} = \log_n a - \log_n b \\
 \log_n a = b \implies a = n^b & \log_n a = \log_n b \implies a = b \\
 n^{\log_n x} = x & \\
 \log_n n^x = x &
 \end{array}$$